

Peripheral Requirements for a 3-Phase EV Inverter using TMS570LS0714APZQQ1

For a 3-phase EV inverter implementing Field-Oriented Control (FOC), the required peripherals on the TMS570LS0714APZQQ1 are summarized below.

1. Magnetic Encoder (AS5047P)

The AS5047P communicates via the SPI interface.

Signal	MCU Peripheral	Pins
SCK	SPI Clock	1
MOSI	SPI SIMO	1
MISO	SPI SOMI	1
CS	SPI Chip Select	1
Total		4 Pins

Table 1: SPI Interface for AS5047P

2. ADC Inputs

The inverter requires four ADC channels.

Signal	Pins
DC Link Voltage	1 ADC
Phase A Current (I_a)	1 ADC
Phase B Current (I_b)	1 ADC
Phase C Current (I_c)	1 ADC
Total	4 ADC Pins

Table 2: Required ADC Channels

3. CAN Bus

A CAN transceiver interfaces with the MCU using two pins.

Signal	MCU Pin
CAN_TX	DCANxTX
CAN_RX	DCANxRX
Total	2 Pins

Table 3: CAN Interface

4. PWM Outputs

A three-phase inverter requires six PWM outputs.

Gate	PWM Output
Phase A High	ePWM1A
Phase A Low	ePWM1B
Phase B High	ePWM2A
Phase B Low	ePWM2B
Phase C High	ePWM3A
Phase C Low	ePWM3B
Total	6 PWM Pins

Table 4: PWM Outputs for Three-Phase Inverter

Total MCU Pins Used

Function	Pins
SPI Encoder	4
ADC Inputs	4
CAN Bus	2
PWM Outputs	6
Total	16 Pins

Table 5: Summary of Required MCU Pins

Optional GPIOs

Additional GPIOs commonly used in a practical EV inverter are listed below.

Function	Pins
Gate Driver Enable	1
Gate Driver Fault Input	1
Brake Input	1
Pre-charge Relay Control	1
Main Contactor Control	1
Status LEDs	2
UART Debug (TX/RX)	2
Total	9 Pins

Table 6: Optional GPIO Requirements

Suggested Peripheral Allocation

Peripheral	Function
ePWM1A/B	Phase A High/Low
ePWM2A/B	Phase B High/Low
ePWM3A/B	Phase C High/Low
MibSPI1	AS5047P Encoder
MibADC1	DC Link Voltage
MibADC1	Phase A Current
MibADC1	Phase B Current
MibADC1	Phase C Current
DCAN1	CAN Transceiver
GPIO	Gate Driver Enable
GPIO	Gate Driver Fault Input

Table 7: Recommended Peripheral Assignment